

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix} \begin{bmatrix} \square & * & * & * & * \\ 0 & \square & * & * & * \\ 0 & 0 & \square & * & * \\ 0 & 0 & 0 & \square & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\quad \quad \quad L \quad \quad \quad U$

Figure 1: An LU factorization

The matrix L is invertible and is called a unit lower triangular matrix.

Usefulness of LU factorization

When $A=LU$, the equation $Ax=b$ can be written as $L(Ux)=b$. Writing y for Ux , we can find x by solving the pair of equations

$$\begin{cases} Ly=b \\ Ux=y \end{cases} \quad (2)$$

First solve $Ly=b$ for y , and then solve $Ux=y$ for x . Each equation is easy to solve because L and U are triangular.

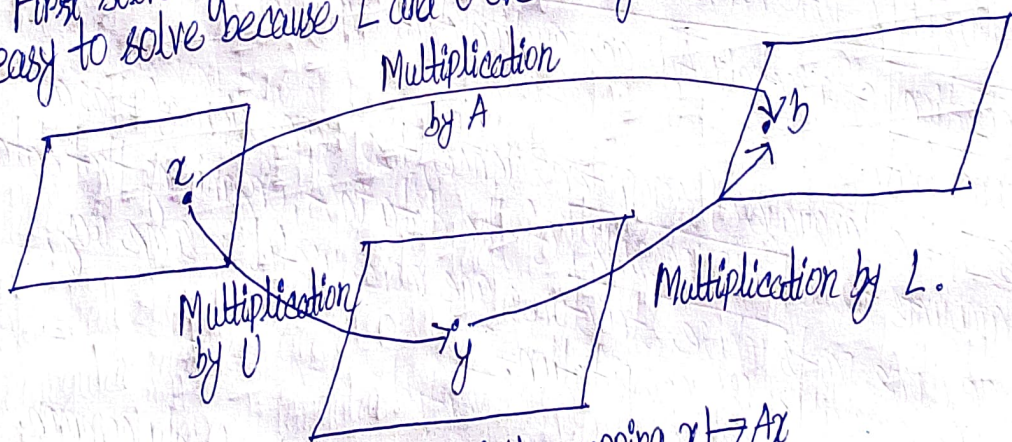


FIGURE 2 Factorization of the mapping $x \mapsto Ax$

Example 1: It can be verified that

$$A = \begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & -5 & 1 & 0 \\ -3 & 8 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} = LU$$

Use this LU factorization of A to solve $Ax=b$, where $b = \begin{bmatrix} -9 \\ 5 \\ 7 \\ 11 \end{bmatrix}$

Solution:

$$Ly=b \Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & -5 & 1 & 0 \\ -3 & 8 & 3 & 1 \end{bmatrix} y = \begin{bmatrix} -9 \\ 5 \\ 7 \\ 11 \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$